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Rejuvenating a Development Strategy for Low-Resistivity Pay Formation in UAE Onshore Oil Field: Restoring Performance and Improving Reservoir Health

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Abstract

Redevelopment of mature oil fields is crucial as operators face declining production, pressure depletion, and sweep inefficiencies. Traditional recovery methods often fall short, prompting the need for innovative, cost-effective solutions to extend field life and unlock bypassed reserves. Field redevelopment enhances reservoir management and supports long-term recovery without major infrastructure changes. This work focuses on redeveloping Reservoir-X, a mature onshore oil field in the UAE, aiming to restore production, improve sweep efficiency, and address pressure sink zones.

The redevelopment strategy integrated advanced reservoir characterization, dynamic simulation, and performance diagnostics to identify pressure sink zones and areas of poor sweep efficiency. New-generation logging tools for low-resistivity pay formations enhanced reservoir description and improved pay zone identification. A pattern-based water injection system was developed to optimize injector producer alignment and spacing, validated through simulation modeling accounting for reservoir heterogeneities. Digital dashboards and real-time monitoring systems were implemented to track well performance and injection efficiency, enabling proactive management and continuous optimization. Additionally, future deployment of Enhanced Oil Recovery (EOR) techniques, including miscible WAG and Disperse Particle Gel (DPG) technologies, was investigated.

The shift from peripheral to pattern-based water injection delivered measurable improvements in reservoir performance. Both areal and vertical sweep efficiencies showed significant enhancement, resulting in more uniform fluid displacement and reduced bypassed oil. Pressure support was effectively restored across previously depleted and underperforming areas, revitalizing overall reservoir energy. Dynamic simulation results demonstrated a substantial uplift in recovery factors, attributed to improved volumetric sweep and a more balanced reservoir pressure profile.

These advancements enabled the field to meet its production targets, while simultaneously extending the productive life of the asset. Reservoir health was restored and estimated ultimate oil recovery increased by more than 10%, optimizing long-term field performance. Additionally, the strategic optimization of existing

infrastructure minimized capital and operational expenditures, making the redevelopment approach both technically robust and economically attractive. Future Enhanced Oil Recovery (EOR) opportunities were evaluated, including Miscible Gas Injection and Disperse Particle Gel (DPG) technologies, with the goal of increasing the recovery factor by an additional 5%, sustaining the production plateau, and achieving a positive net present value (NPV).

This redevelopment of Reservoir-X presents a novel application of pattern-based injection in a mature, low-resistivity reservoir, using existing infrastructure to improve VRR, sweep efficiency and reservoir health. The integration of advanced logging tools and digital monitoring systems enabled precise reservoir management without major infrastructure expansion. This cost-effective, scalable approach sets a practical model for similar mature field redevelopment efforts.

Introduction

The redevelopment of mature oil fields has become an essential strategy in sustaining global oil production as conventional resources face natural decline. Globally, more than 70% of oil production comes from mature fields, many of which are already well into their secondary or tertiary recovery stages (IEA, 2018). These fields often suffer from challenges such as reservoir pressure depletion, water breakthrough, coning, and sweep inefficiencies, which collectively contribute to a steep decline in production rates over time (Muggeridge et al., 2014). As a result, operators are compelled to adopt innovative, integrated approaches to prolong the economic viability of these aging assets.

Traditional recovery methods such as primary depletion and conventional waterflooding are often insufficient for extracting the remaining oil, particularly in heterogeneous or compartmentalized reservoirs. A significant portion of original oil in place (OOIP) remains unrecovered even after decades of production. Studies have shown that global average recovery factors for conventional oil fields range between 20–40% under primary and secondary recovery, leaving substantial volumes of hydrocarbons untapped (Al-Mutairi & Kokal, 2011). Therefore, redevelopment strategies aimed at enhancing reservoir contact and optimizing fluid flow become increasingly important. These may include infill drilling, horizontal wells, selective recompletions, artificial lift optimization, and the application of advanced reservoir surveillance and modeling technologies.

Field redevelopment offers a cost-effective means to maximize oil recovery without the need for major infrastructure upgrades. By leveraging existing facilities and integrating real-time data from modern monitoring systems, operators can significantly improve recovery while maintaining operational efficiency (Satter et al., 2016). Moreover, redevelopment is aligned with broader energy transition goals, as it enables more efficient resource utilization and reduces the carbon intensity per barrel produced by deferring the need for new field developments.

This work focuses on the redevelopment of Reservoir-X, a mature onshore oil field located in the United Arab Emirates (UAE). Over its production life, Reservoir-X has experienced notable declines in reservoir pressure and production rates, attributed to historical waterflooding inefficiencies and the emergence of pressure sink zones. The redevelopment plan aims to address these issues through targeted interventions that enhance sweep efficiency, restore pressure support, and access bypassed reserves. Through advanced reservoir characterization, infill well placement, and optimized well pattern design, this study presents a practical framework for revitalizing production in mature fields while minimizing additional capital expenditure.

Field Development Planning (FDP)

Field Development Planning (FDP) is a comprehensive, multidisciplinary process designed to outline how an oil or gas field will be developed and produced throughout its economic life. Its primary goal is to maximize hydrocarbon recovery while balancing technical viability, economic returns, operational risks,

and environmental responsibility. Rather than being a static plan, FDP is a dynamic and iterative framework that evolves with new data, reservoir behavior changes, and market fluctuations (Kaiser and Pulsipher, 2004).

The FDP process typically progresses through several phases, starting with concept selection and advancing through to detailed engineering and execution. It requires the integration of diverse elements such as subsurface characterization, well design, production forecasting, facility planning, economic evaluation, and regulatory compliance.

At the core of effective FDP lies robust subsurface evaluation. This involves building static models from geological and petrophysical interpretations, complemented by dynamic models developed through reservoir simulation. These models are critical in estimating original hydrocarbons in place (OOIP/OGIP), projecting recovery factors, and assessing alternative development strategies (Begg et al., 2001). During early stages, several development options are evaluated for instance, vertical versus horizontal wells, waterflooding versus gas injection, or conventional versus enhanced oil recovery (EOR) techniques. These options are ranked using performance indicators such as Net Present Value (NPV), Internal Rate of Return (IRR), payout time, and recovery efficiency. Well planning is a central pillar of FDP. It involves determining optimal well trajectories, spacing, and injector-producer configurations to maximize reservoir contact and ensure efficient drainage. For complex or mature reservoirs, advanced tools such as 3D geocellular models, streamline simulations, and geosteering technologies help refine well placement (Jafarpour and He, 2009). Offshore or remote developments must also consider logistical challenges, rig scheduling, and surface facility constraints. Cutting-edge technologies like intelligent completions, multilateral wells, and extended-reach drilling are increasingly incorporated to improve reservoir access and minimize environmental footprint.

Surface facility planning is seamlessly integrated with reservoir planning to ensure infrastructure such as separators, pipelines, compressors, and injection systems is appropriately designed and scaled. Key facility-related decisions include selecting between modular and centralized systems, determining fluid handling capacity, designing export routes, and managing power needs. Additionally, facilities must support regulatory compliance and be adaptable for future expansions or debottlenecking efforts, with considerations for gas reinjection and flaring minimization.

A critical component of FDP is economic evaluation and risk analysis. This involves preparing detailed cost breakdowns, forecasting production profiles, and conducting sensitivity analyses to understand project viability under various market and reservoir scenarios (Begg et al., 2001). Tools such as decision trees, Monte Carlo simulations, and probabilistic modeling help quantify uncertainty and guide capital investment decisions. Value of Information (VoI) analysis may also be employed to assess the benefits of acquiring additional data such as from appraisal wells or seismic surveys before committing to full-scale development.

In modern planning environments, FDPs are increasingly supported by digital technologies and data-driven approaches. These include integrated asset models, machine learning for scenario optimization, cloud-based collaboration tools, and real-time monitoring from smart wells and digital twins. Such advancements enable adaptive field development strategies that evolve alongside new operational insights during early production phases.

Ultimately, Field Development Planning is a foundational process that shapes the technical, economic, and operational path of a hydrocarbon asset. Its effectiveness depends on high-quality reservoir data, close cross-disciplinary collaboration, rigorous economic assessment, and an agile approach to decision-making. When well-executed, FDP enhances recovery, improves financial performance, mitigates risk, and aligns development with long-term field goals.

Fig. 1 presents standard industry workflows and economic analysis logic (Kaiser and Pulsipher, 2004; Begg et al., 2001), with four key visualizations highlighting critical FDP elements:

1. FDP Workflow Progress – Outlines key planning stages from reservoir characterization to execution.

2. Production Forecast Scenarios – Compares output projections across different development strategies.
3. Recovery Factor vs. Well Spacing – Illustrates the impact of well spacing on recovery efficiency.
4. NPV Sensitivity Analysis – Examines how key variables influence project economic performance.

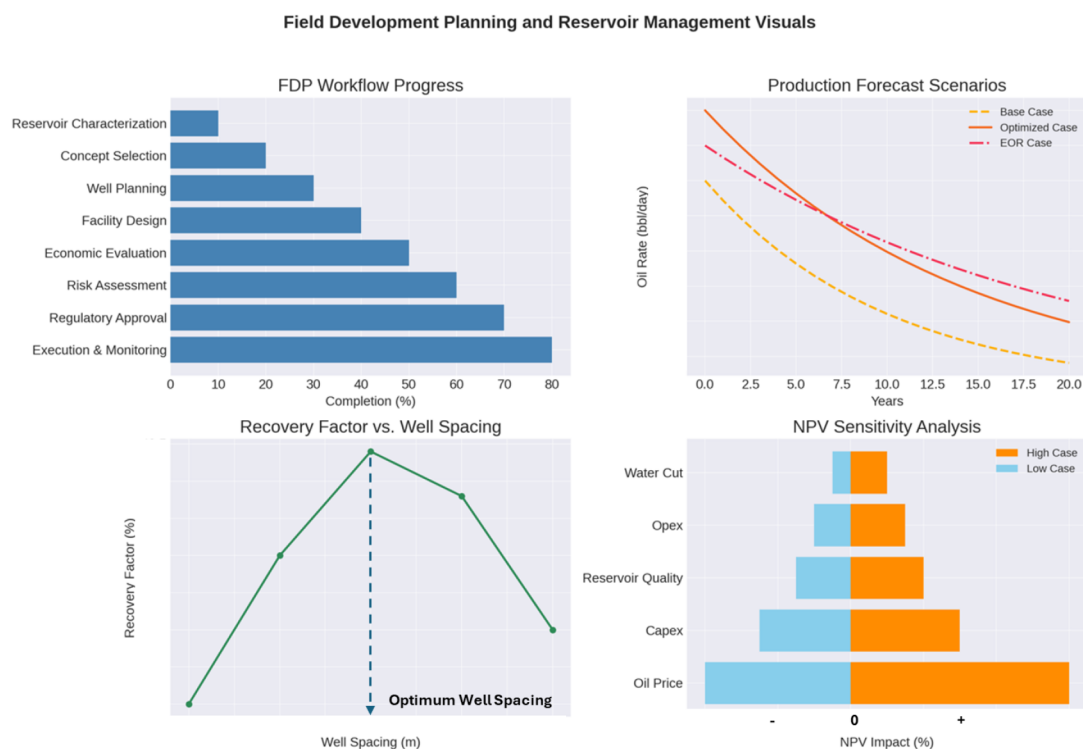


Figure 1—Standard Field Development Planning Workflows and Economic Analysis Logic

Water Injection Patterns

Water injection is a key secondary recovery method employed to maintain reservoir pressure and improve oil displacement toward production wells. The effectiveness of water injection depends heavily on the selection and design of the injection pattern, which governs the flow paths of both injected water and displaced oil within the reservoir. Properly designed patterns enhance volumetric sweep efficiency, reduce oil bypassing, and mitigate early water breakthrough, thereby maximizing recovery and minimizing production challenges such as excessive water cut.

Injection patterns can be broadly classified into regular patterns (geometrically arranged) and irregular patterns (customized to fit reservoir heterogeneities or development constraints). Among the most common regular patterns are the five-spot, nine-spot, and inverted five-spot. In a five-spot pattern, an injector is surrounded by four producers forming a square or diamond grid, creating a balanced radial displacement of oil. This pattern offers good areal coverage and is typically used in uniform reservoirs with minimal heterogeneity. The inverted five-spot rearranges the positions placing a producer at the center and injectors at the corners which can offer improved sweep in certain stratified or fractured reservoirs by better controlling the direction of water movement.

The nine-spot pattern, consisting of eight injectors surrounding a central producer, increases areal density and provides more injection support per producer. It is suitable for high-permeability or high-oil-saturation reservoirs where aggressive injection is needed.

Line drive patterns, where injectors and producers are placed in alternating rows, are ideal for elongated reservoirs or where directional control of the flood front is desired. This pattern creates a piston-like displacement, which can be advantageous in layered or anisotropic formations, especially when aligned with the principal permeability direction. Peripheral water injection, often applied in carbonate or dome-shaped

reservoirs, involves locating injectors along the reservoir boundary to force oil inward toward producers. It is particularly effective in maintaining pressure throughout the field and controlling external edge-water encroachment.

The selection of an appropriate injection pattern is governed by multiple geological and operational factors, including reservoir geometry, fluid properties, well spacing, permeability distribution, fault presence, and existing well infrastructure. In heterogeneous or compartmentalized reservoirs, irregular or hybrid patterns often determined using reservoir simulation and dynamic modeling can provide better adaptability to local flow conditions. For example, in fields with natural fractures or high-permeability streaks, standard patterns may lead to uneven displacement, so adaptive pattern modifications or selective zonal injection strategies may be employed to improve conformance and contact efficiency. Fig. 2 illustrates different types of well patterns.

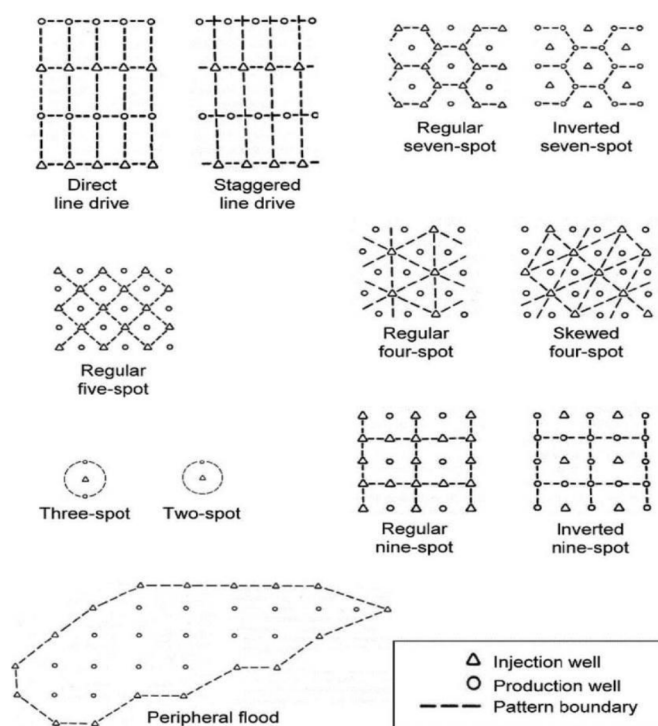


Figure 2—Different Types of Well Patterns (Lahkar, Nitin 2019)

Modern field redevelopment often leverages real-time data acquisition (e.g., from pressure sensors, tracers, or distributed temperature sensing) and reservoir simulation tools to monitor injection efficiency and adjust patterns dynamically. Additionally, technologies like inflow control devices (ICDs) and intelligent completion systems have further improved the management of injection profiles, allowing for refined control over vertical and areal sweep. Ultimately, the optimization of water injection patterns plays a vital role in maximizing oil recovery, minimizing water handling costs, and extending the productive life of mature oil fields.

Summary for the Common Injection Patterns

1. Five-Spot Pattern

In this widely used pattern, one injection well is placed in the center of a square surrounded by four production wells at the corners. This configuration offers symmetrical displacement and high areal coverage, making it effective in homogeneous reservoirs.

2. Inverted Five-Spot Pattern

This is the reverse of the five-spot, with a central producer and injectors at the corners. It's advantageous in situations requiring tighter production control and is often used in pilot tests or small-scale redevelopments.

3. **Nine-Spot Pattern**

Consisting of a single producer surrounded by eight injectors, this dense layout provides greater control over injection volumes and improves sweep in more complex formations, especially those with varying permeability.

4. **Line Drive Pattern**

A row of injection wells faces a parallel row of producers. This is ideal for elongated or fault-bounded reservoirs, as it creates a linear displacement front, mimicking a piston-like drive.

5. **Peripheral (Edge) Injection**

In this pattern, injectors are located along the reservoir's periphery to push oil inward toward centrally located producers. This method is common in dome- or anticline-shaped reservoirs and is effective for maintaining uniform pressure across a wide area.

Reservoir-X Redevelopment Case Study

Reservoir Pressure History

The reservoir has been experiencing a gradual pressure decline in the central area due to peripheral water injection and the significant distance between injectors and producers, which has limited the efficiency of pressure support in the middle of the field. Despite these challenges, pressure levels have been successfully maintained over the past few years through ongoing management strategies and operational adjustments. However, to enhance pressure distribution and improve sweep efficiency, the reservoir development strategy has been revised, shifting from the current peripheral injection scheme to a more effective pattern water injection approach. This re-planning aims to optimize water flooding, ensure better reservoir contact, and ultimately sustain production rates over the long term.

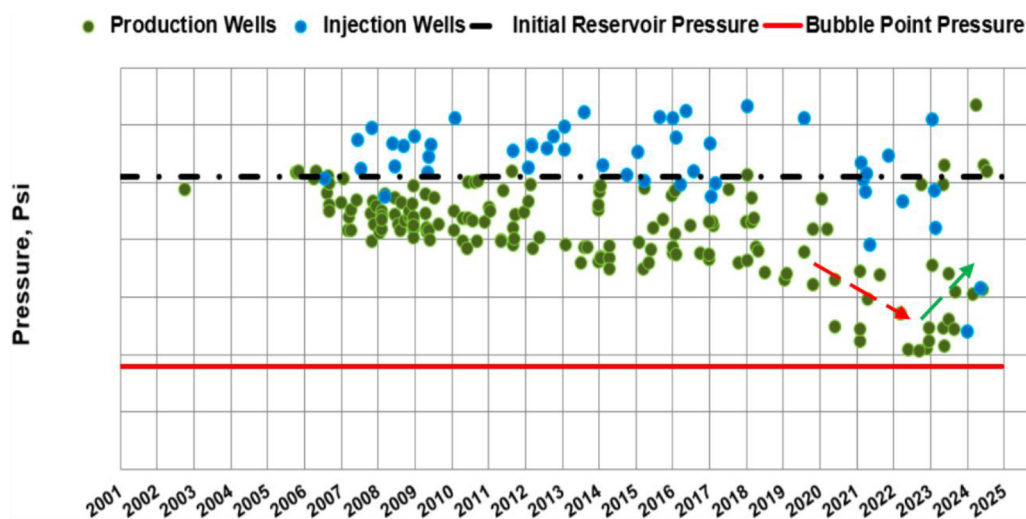


Figure 3—Reservoir Pressure History

New Field Development Concept

The field has been facing several subsurface challenges that have hindered optimal performance and production efficiency. One of the primary issues stems from the Low Resistivity Pay (LRP), which, despite being located within the oil column, often exhibits water-bearing characteristics, complicating interpretation and well placement. The LRP phenomena is characterized by stratigraphic interval with $R_t \log$ value of < 1

ohm.meter and commonly developed in the middle of the subzone. It corresponds to wackestone-floatstone lithology with abundant micritized matrix with micropores, high porosity >20%, and high salinity > 170000 ppm (Salahuddin, 2015).

Additionally, the High Permeability Streaks (HPS) present within the reservoir have contributed to rapid water advancement, leading to early water breakthrough in certain wells. The HPS corresponds to grainstone rich interval that underwent leaching diagenetic process. It is 2-3 feet thick and enveloped the LRP at the top and at the bottom. The HPS is identified and validated through core observation, thin section description, MICP, NMR, and production logging tools (Salahuddin, 2016). The HPS and LRP are two distinct rock types, each characterized by unique ranges of porosity, permeability, and capillary pressure. Stratigraphically, the HPS are located at the top and base of the middle subzone, while the LRP is predominantly situated in the central portion of the same interval. To accurately capture their spatial distribution within the 3D geological model (Fig. 4), a combination of property modeling techniques was employed, including Proportion Matrix: to define the relative occurrence within each subzone; Vertical Proportion Curves (VPCs): to constrain the vertical distribution trends; Lateral Probability Maps: to guide the horizontal continuity and spatial probability of rock type presence.

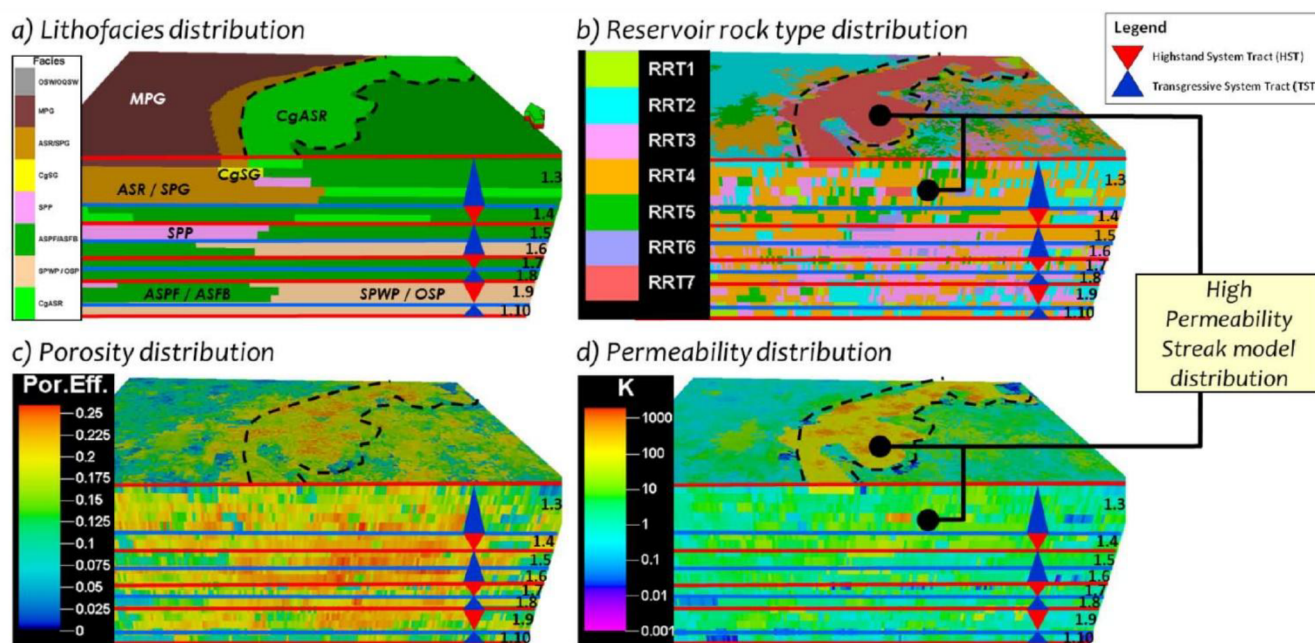


Figure 4—3D Geological Models Realization with a Combination of Property Modeling Techniques (Salahuddin, 2016)

These modeling techniques enable a probabilistic understanding of the 3D distribution of HPS and LRP, which is crucial for predicting their impact on reservoir dynamic behavior. This understanding directly dictates well placement strategies, ensuring that wells are optimally located to maximize reservoir contact and production efficiency. These geological complexities were not adequately captured in the previous Field Development Plan (FDP), as the older reservoir model failed to fully represent these heterogeneities, resulting in an unintended pressure sink developing in the central portion of the field.

To address these issues, a comprehensive resolution plan has been established. The latest subsurface study has successfully incorporated the effects of both LRP and HPS into an updated reservoir model, providing a more accurate representation of the field. Building on these insights, a new FDP has been rolled out, focusing on pattern water injection to restore pressure across the reservoir and improve sweep efficiency.

Maximizing reservoir exposure and contact in such case requires accurately targeting a thin zone while addressing structural uncertainties. Due to the low resistivity contrast between the optimal zone and

its surroundings, an advanced mapping tool is necessary to provide clear insight into the environment. The UDR-DTB (Ultra-Deep Resistivity –Distance to Boundary) tool was placed near the bit to map the resistivity distribution, followed by a density porosity tool to confirm reservoir properties. The recent implementation demonstrated the effectiveness of this technology in improving hydrocarbon recovery and reducing operational risks in such complex environment. This proactive approach reduces uncertainties and supports strategic planning for future wells, opening promising opportunities (Salahuddin, 2025).

As part of this plan, well placement strategies have been further optimized through the deployment of UDR-DTB (technology, which allows for precise geosteering in challenging thin pay zones. This advanced approach has already proven successful, with four wells meeting the technical requirements and more than 5,000 feet of measured depth (MD) of horizontal sections placed accurately within the preferred reservoir intervals. Together, these initiatives are expected to mitigate prior development challenges, enhance reservoir management, and unlock significant additional production potential. Fig. 5 illustrates New Development Concept for Reservoir-X.

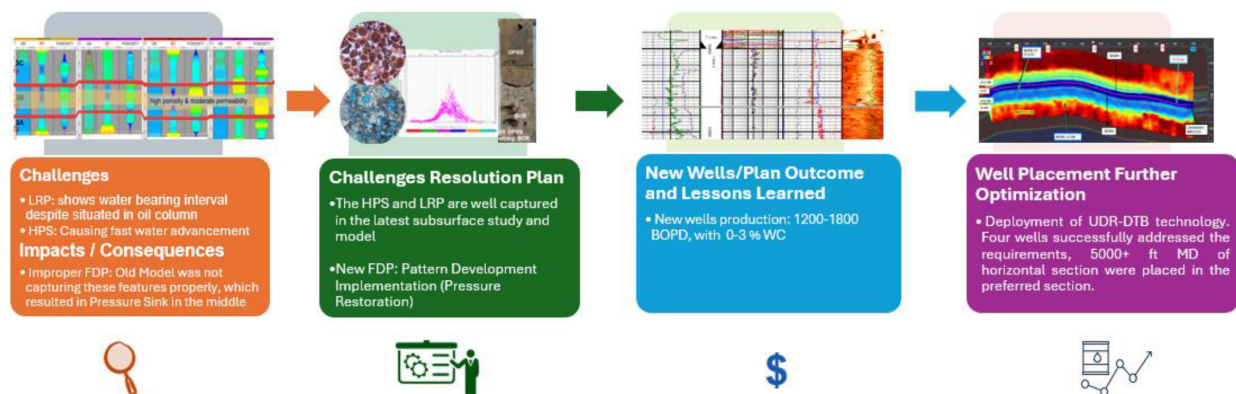


Figure 5—New Development Concept for Reservoir-X

Re-developments Wells from Peripheral to Pattern

The implementation of the new Field Development Plan (FDP) is currently underway and is expected to significantly address the underperformance observed in the Reservoir-X. The revised FDP involves a strategic shift from a peripheral injection approach to a pattern injection system, aimed at providing more effective pressure support in the central area of the field, thereby enhancing overall reservoir sweep efficiency.

The transition from peripheral to pattern injection was guided by comprehensive reservoir simulation and sensitivity analyses to optimize pressure support and recovery efficiency. Various pattern configurations and well spacings were tested to balance sweep efficiency with drilling costs, with results indicating that tighter spacing in high-permeability zones maximized recovery while minimizing early water breakthrough. Sensitivities on field-level injection rates were also evaluated to define an optimal range that sustains reservoir pressure without causing injectivity issues or bypassed oil. Infill well tie-in strategies were incorporated into the development plan, ensuring alignment with existing surface facilities to minimize CAPEX while accelerating production ramp-up. Furthermore, dynamic simulation and history-matching workflows enabled the identification of "sweet spots" with higher hydrocarbon saturation and favorable rock properties, which were prioritized for early drilling and injection placement. This integrated approach ensured that the re-development plan was robust, technically justified, and economically optimized.

A key highlight of this plan is the potential to unlock production gains of up to 6 million stock tank barrels (MMSTB) per well from the low resistivity pay reservoir. This achievement is being driven by the innovative deployment of the UDR-DTB tool in addition to the new wells patterns and spacing. By bringing forward the idea, thoroughly investigating, evaluating, and ultimately applying this advanced technology,

the development team is paving the way for improved reservoir exploitation, reduced drilling risks, and a step-change in production performance from this challenging low-resistivity formations. Fig. 6 shows the shift from peripheral to pattern field development in Reservoir-X.

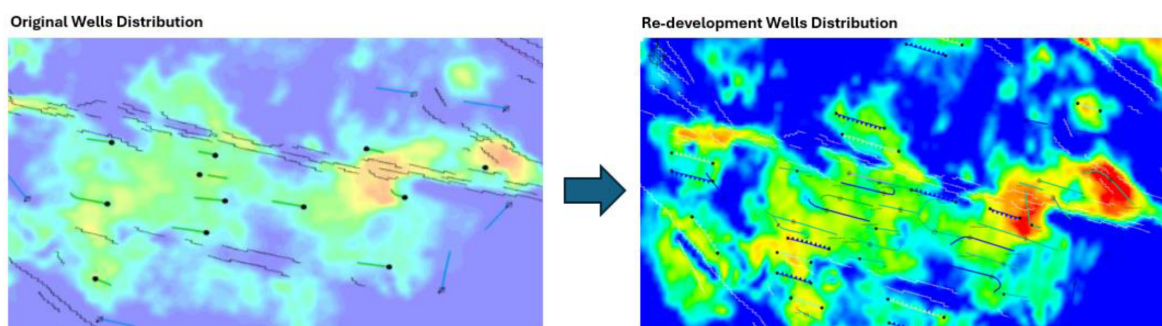


Figure 6—Shift from Peripheral to Pattern Water Injection FDP in Reservoir-X

Water Production Control For Further Well Performance Optimization

As part of well performance optimization to manage excessive water production and improve reservoir performance, advanced lower completion strategies have been implemented incorporating Inflow Control Devices (ICDs) and Selective Sliding Sleeves (SSDs) through detailed reservoir simulation and sensitivity analysis to ensure effective zonal flow control and water management. Multiple ICD designs were tested case by case to balance inflow across heterogeneous reservoir layers, with placement criteria based on permeability contrast, expected pressure gradients, and fluid contact proximity. Sensitivity runs assessed how different ICD nozzle sizes and pressure-drop settings influenced inflow distribution, ultimately selecting configurations that minimized early water breakthrough while enhancing sweep efficiency. SSDs were strategically placed in intervals with higher uncertainty or greater water production risk, providing operational flexibility to selectively choke back or shut off unwanted zones. The placement strategy prioritized zones with favorable reservoir quality and oil saturation, while maintaining alignment with the overall injection pattern and infill well tie-in strategy. This zonal control optimization ensured that the completion design maximized recovery while prolonging well productivity and reducing water handling costs.

Fig. 7 presents an example from Well-X1 that was originally designed to be completed as an open-hole oil producer like the old wells, but due to reservoir complexities and production challenges, a retrofitted Inflow Control Device (ICD) design with strategically placed blank sections was implemented to specifically manage and balance inflow from the high-permeability zones, which were contributing to early water breakthrough. This modification proved highly effective, as the post-installation performance of the well demonstrated a reduction of more than 60% in water production, alongside a notable stabilization of the oil flow rate. When compared to a nearby well completed without ICD, Well-X1 showcased significantly improved production sustainability, validating the effectiveness of ICD deployment in mitigating water production issues and optimizing reservoir performance.

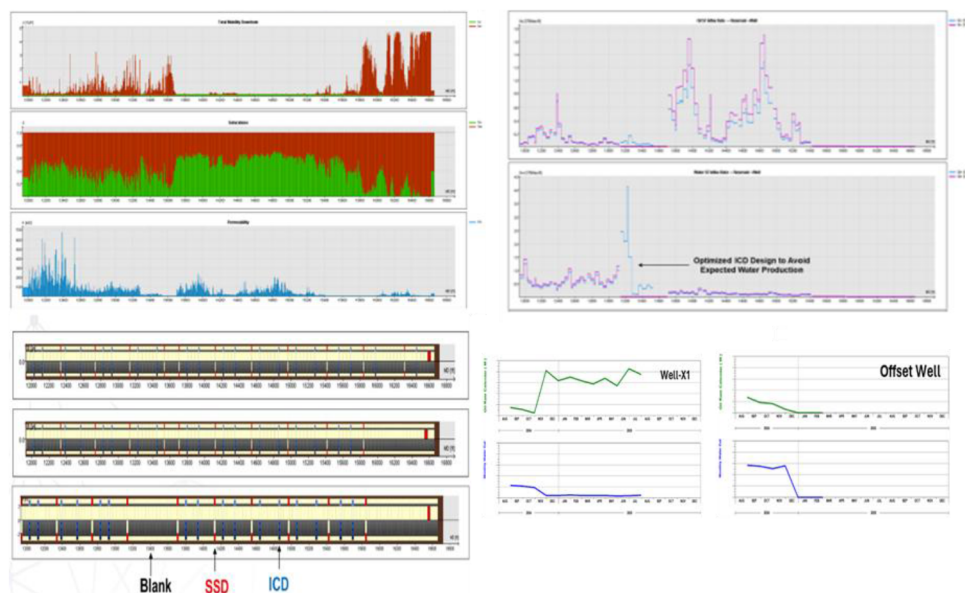


Figure 7—Well-X1 Example of Using Lower Completion with ICDs and SSDs for Effective Water Production Control

Role of Digitilization in Wells Monitoring

Using digital platforms provide a powerful and centralized solution for monitoring critical well parameters, including Wellhead Pressure (WHP), Wellhead Temperature (WHT), Gas Lift Rate, Injection Rate, and other key production indicators. By integrating real-time data from field sensors and SCADA systems, the digital platform allows to visualize and track well performance through intuitive dashboards and trend displays.

The digital platform has played a vital role in diagnosing and resolving well performance issues in real time by continuously monitoring producers and injectors under actual flowing conditions. This enhanced visibility improves situational awareness, enabling engineers to quickly recognize anomalies such as pressure drops, temperature fluctuations, or irregular gas-lift and injection responses.

In practice, the platform has flags abnormal well behaviors that signals potential operational problems or near-wellbore damage, prompting timely interventions that helps restoring and sustaining productivity. On the injection side, continuous tracking of rates and pressures has made it possible to identify injectors performing below target due to plugging or pressure constraints, allowing for rapid corrective actions that improved injectivity.

Real-time monitoring has also highlighted imbalances between injectors and nearby producers, supporting quick adjustments in injection allocation to re-establish uniform sweep efficiency. These capabilities ensure stable reservoir pressure support, reduced downtime, and more consistent production targets without reliance on costly, reactive interventions.

In addition, the platform's ability to analyze historical data provides further insights for diagnosing recurring issues, optimizing operating conditions, and guiding proactive reservoir management strategies.

Overall, the use of this digital platform has strengthened operational efficiency, enabled data-driven decision-making, and minimized downtime by delivering timely, actionable insights into well performance.

Fig. 8 shows an example of real time monitring of oil producers from the digital platform with the main operating parameters.

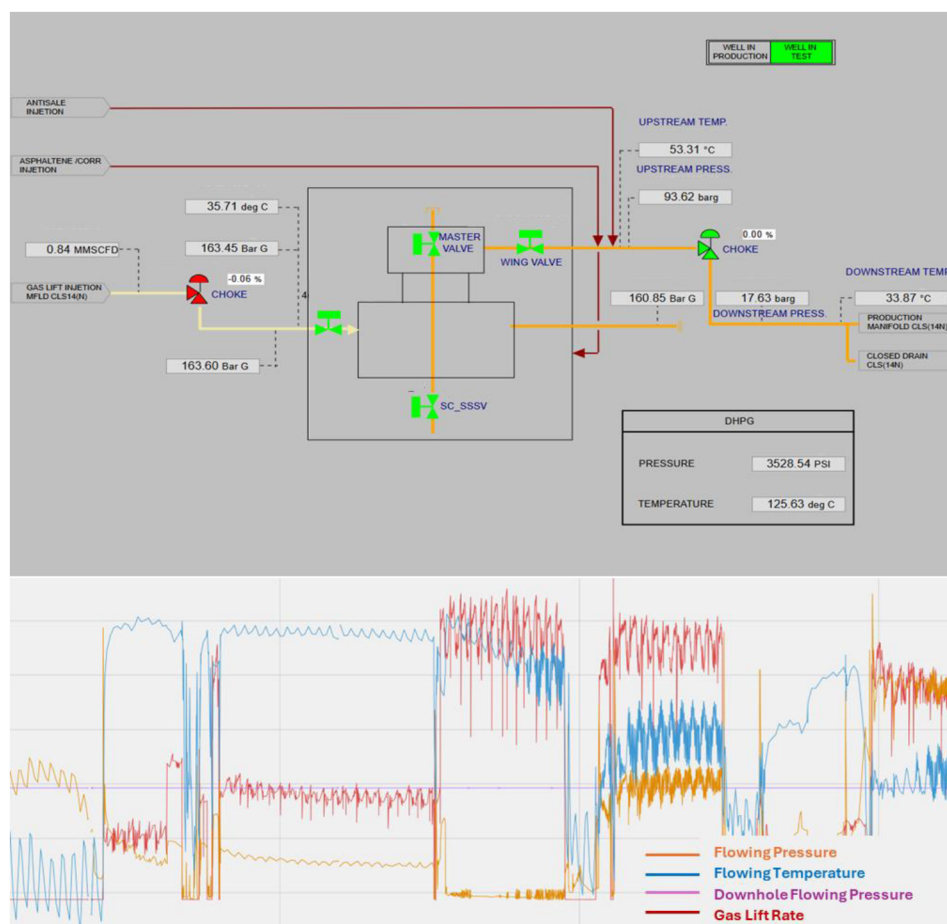


Figure 8—Example of Real Time Monitoring of Oil Producers

Additional Gain from the New Development

There are promising additional opportunities to further enhance field recovery, particularly through the planned implementation of a Disperse Particle Gel (DPG) pilot project. DPG consists of nano- to micron-sized soft gel particles that demonstrate excellent stability under harsh reservoir conditions, withstanding temperatures up to 150°C and salinity levels as high as 250,000 mg/L (Siqing, 2024). Recognizing its potential, an EOR pilot program is scheduled to commence in 2025, targeting two wells currently experiencing high water cut, an issue confirmed to be linked to injection water migration through tracer studies.

Extensive laboratory analyses have already been conducted, indicating that DPG injection could deliver up to a 15% increase in recovery factor by improving conformance control and blocking high-permeability water channels. Furthermore, a numerical simulation study supporting the pilot proposal estimates an additional 220–300 Mbbbl of oil recovery underscoring the potential economic impact of the technology.

Once the pilot is underway, critical performance indicators such as oil production rate, water cut (WC), and subsurface diagnostics will be closely monitored to assess the effectiveness of the applied strategy and provide a clear basis for full-field implementation decisions. Monitoring production rates and WC trends will help quantify incremental oil recovery, identify potential improvements in sweep efficiency, and detect early signs of water breakthrough or channeling. In parallel, subsurface diagnostics obtained through surveillance tools and periodic well tests will deliver valuable insights into fluid movement, reservoir pressure distribution, and zonal contributions, ensuring that performance is aligned with simulation forecasts. The integration of these measurements with real-time data analytics will enable continuous evaluation of pilot performance, timely optimization of well operating conditions, and adaptive decision-

making. Ultimately, this systematic monitoring framework will reduce uncertainty, validate reservoir models, and ensure that any scale-up to full-field deployment is both technically justified and economically viable.

Conclusion

The Field Development Plan (FDP) for Reservoir-X has undergone a significant redevelopment, transitioning from a peripheral injection approach to a pattern injection strategy to better maintain pressure support in the central area and enhance reservoir management. Implementation of this new FDP is already underway, with recently drilled wells delivering positive outcomes that help mitigate the reservoir's historical underperformance.

A key milestone in the redevelopment effort has been the deployment of the Ultra-Deep Resistivity–Directional Thin Bed (UDR-DTB) tool for the first time in the field, addressing the challenges posed by low-resistivity pay (LRP) and high-permeability streaks (HPS). Following extensive investigation, evaluation, and execution, this advanced technology has enabled precise well placement within 5–7 ft thin pay zones, unlocking up to 6 million stock tank barrels (MMSTB) of additional oil per well, complementing the optimized injection pattern and well spacing.

Additional redevelopment measures, including advanced lower completion strategies and digital well monitoring, further enhance well performance and provide real-time surveillance for proactive optimization.

Beyond the FDP, opportunities to maximize recovery are being actively evaluated, with full-field Enhanced Oil Recovery (EOR) options under assessment, including miscible EOR techniques for further opportunity of raising the recovery factor (RF). In parallel, an EOR pilot for Disperse Particle Gel (DPG) injection is being implemented to improve conformance and reduce excessive water production.

Collectively, these redevelopment initiatives represent a comprehensive strategy to improve reservoir management, enhance oil recovery, and secure long-term production sustainability.

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